In [2]:
```
#Import pandas library for data analysis
import pandas as pd

#load dataser from CSV file into DataFrame
df = pd.read_csv("ids.csv")
```

In [3]:
```
#Display last 50 rows of dataset
print("Last 50 records")
df.tail(50)
```

```
Last 50 records
```

Out[3]:

	Unnamed: 0	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Max	Fwd Packet Length Min	Fwd Packet Length Mean	...	min_seg_size_forward	Active Mean	Active Std	Active Max	Active Min	Idle Mean	Idle Std	Idle Max	Idle M
488065	488065	53	70591	2	2	66	194	33	33	33.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488066	488066	443	5572560	8	6	386	5205	205	0	48.250000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488067	488067	3013	49	2	0	4	0	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488068	488068	53	122130	2	2	80	272	40	40	40.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488069	488069	53	98106274	2	2	114	211	58	56	57.000000	...	20	30943.00000	0.00000	30943	30943	9.800000e+07	0.00000	98000000	980000
488070	488070	53	506	1	1	53	113	53	53	53.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488071	488071	53	3957631	2	2	120	260	79	41	60.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488072	488072	9102	2	2	0	4	0	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488073	488073	2135	111	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488074	488074	1048	49	2	0	4	0	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488075	488075	54240	55	1	1	6	6	6	6	6.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488076	488076	1049	73	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488077	488077	443	4	2	0	12	0	6	6	6.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488078	488078	53	172	2	2	80	112	40	40	40.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488079	488079	443	79063	1	2	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488080	488080	51425	4	2	0	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488081	488081	5120	85	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488082	488082	39930	4	2	0	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488083	488083	443	5871633	7	4	635	168	517	0	90.714286	...	32	150944.00000	0.00000	150944	150944	5.720685e+06	0.00000	5720685	57206
488084	488084	10616	99	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488085	488085	443	115338719	22	18	1317	7539	864	0	59.863636	...	20	84994.45455	93143.49644	365777	56220	9.992959e+06	27637.53213	10000000	99105
488086	488086	60928	52	1	1	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488087	488087	22	1340965	41	42	2728	6634	456	0	66.536585	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488088	488088	443	1142446	78	107	997	227837	517	0	12.782051	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488089	488089	53	67977	2	2	70	130	35	35	35.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488090	488090	53	32534951	2	2	130	297	79	51	65.000000	...	32	475.00000	0.00000	475	475	3.250000e+07	0.00000	32500000	325000
488091	488091	53	31361	2	2	70	198	35	35	35.000000	...	40	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488092	488092	53	228451	2	2	62	240	31	31	31.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488093	488093	53	216	2	2	78	160	39	39	39.000000	...	40	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488094	488094	6302	47740	2	0	6	0	6	0	3.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488095	488095	53	210	2	2	70	130	35	35	35.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488096	488096	443	5322619	5	5	3471	1132	1993	6	694.200000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488097	488097	53	118556	2	2	76	300	38	38	38.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488098	488098	389	69	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488099	488099	5405	72	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488100	488100	53	206842	1	1	48	142	48	48	48.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488101	488101	8402	49	2	0	4	0	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488102	488102	21	72	1	2	6	12	6	6	6.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488103	488103	43118	69	1	1	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488104	488104	53	48546	2	2	70	102	35	35	35.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488105	488105	49848	19	2	2	12	4	6	6	6.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488106	488106	443	31143	4	1	190	0	190	0	47.500000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488107	488107	53	51160	1	1	43	113	43	43	43.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488108	488108	80	23207	2	0	12	0	6	6	6.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488109	488109	80	5149971	3	1	0	0	0	0	0.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488110	488110	53	238	2	2	86	228	43	43	43.000000	...	32	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488111	488111	8899	3	2	0	4	0	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488112	488112	53	250	2	2	82	206	41	41	41.000000	...	20	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	
488113	488113	53	70581019	2	2	117	231	66	51	58.500000	...	32	41306.00000	0.00000	41306	41306	7.030000e+07	0.00000	70300000	703000
488114	488114	5679	121	2	2	4	12	2	2	2.000000	...	24	0.00000	0.00000	0	0	0.000000e+00	0.00000	0	

50 rows × 80 columns

In [4]:
```
#Define required columns for analysis
columns = ['Destination Port', 'Flow Duration', 'Total Fwd Packets', 'Total Backward Packets',
           'Total Length of Fwd Packets', 'Total Length of Bwd Packets', 'Fwd Packet Length Mean',
           'Bwd Packet Length Mean', 'Flow Bytes/s', 'Flow Packets/s', 'Packet Length Mean', 'Packet Length Std',
           'Average Packet Size', 'Active Mean', 'Idle Mean', 'Label']

#Create new dataframe with selected columns
df_new = df[columns].copy()
```

In [5]:
```
#Displays new column in dataframe
print("Columns in new DataFrame:")
print(df_new.columns)
```

```
Columns in new DataFrame:
Index(['Destination Port', 'Flow Duration', 'Total Fwd Packets',
       'Total Backward Packets', 'Total Length of Fwd Packets',
       'Total Length of Bwd Packets', 'Fwd Packet Length Mean',
       'Bwd Packet Length Mean', 'Flow Bytes/s', 'Flow Packets/s',
       'Packet Length Mean', 'Packet Length Std', 'Average Packet Size',
       'Active Mean', 'Idle Mean', 'Label'],
      dtype='object')
```

In [6]:
```
#Print toal number of columns in new DataFrame
print("Total number of columns:", df_new.shape[1])
```

```
Total number of columns: 16
```

In [7]:
```
#Dislay first 5 rows of new Dataframe
df_new.head()
```

Out[7]:

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Mean	Bwd Packet Length Mean	Flow Bytes/s	Flow Packets/s	Packet Length Mean	Packet Length Std	Average Packet Size	Active Mean	Idle Mean	Label
0	53	62171	2	2	78	164	39.00	82.000000	3.892490e+03	64.338679	56.200000	23.552070	70.250000	0.00000	0.0	BENIGN
1	2710	48	2	0	4	0	2.00	0.000000	8.333333e+04	41666.666670	2.000000	0.000000	3.000000	0.00000	0.0	Infiltration
2	443	4792909	5	1	135	46	27.00	46.000000	3.776412e+01	1.251849	32.428571	18.866700	37.833333	0.00000	0.0	BENIGN
3	80	115596470	75	82	342	118155	4.56	1440.914634	1.025092e+03	1.358173	749.981013	946.124811	754.757962	22092.63636	10000000.0	BENIGN
4	2910	14	2	2	4	12	2.00	6.000000	1.142857e+06	285714.285700	3.600000	2.190890	4.500000	0.00000	0.0	BENIGN

In [8]:
```
# Check missing values
df_new.isnull().sum()
```

Out[8]:
```
Destination Port                0
Flow Duration                   0
Total Fwd Packets               0
Total Backward Packets          0
Total Length of Fwd Packets     0
Total Length of Bwd Packets     0
Fwd Packet Length Mean          0
Bwd Packet Length Mean          0
Flow Bytes/s                   36
Flow Packets/s                  0
Packet Length Mean              0
Packet Length Std               0
Average Packet Size             0
Active Mean                     0
Idle Mean                       0
Label                           0
dtype: int64
```

In [9]:
```
# Remove rows with missing values
df_new = df_new.dropna()
```

In [10]:
```
# Check again to confirm removal
df_new.isnull().sum()
```

Out[10]:
```
Destination Port               0
Flow Duration                  0
Total Fwd Packets              0
Total Backward Packets         0
Total Length of Fwd Packets    0
Total Length of Bwd Packets    0
Fwd Packet Length Mean         0
Bwd Packet Length Mean         0
Flow Bytes/s                   0
Flow Packets/s                 0
Packet Length Mean             0
Packet Length Std              0
Average Packet Size            0
Active Mean                    0
Idle Mean                      0
Label                          0
dtype: int64
```

In [11]:
```
# Check total number of duplicate rows
duplicates = df_new.duplicated().sum()

#Display duplicate count
print("Total duplicate records:", duplicates)
```

```
Total duplicate records: 139389
```

In [12]:
```
# Total number of columns
print("Total columns:", df_new.shape[1])
```

```
Total columns: 16
```

In [13]:
```
#Display all Column names
print("Column names:")
print(df_new.columns)
```

```
Column names:
Index(['Destination Port', 'Flow Duration', 'Total Fwd Packets',
       'Total Backward Packets', 'Total Length of Fwd Packets',
       'Total Length of Bwd Packets', 'Fwd Packet Length Mean',
       'Bwd Packet Length Mean', 'Flow Bytes/s', 'Flow Packets/s',
       'Packet Length Mean', 'Packet Length Std', 'Average Packet Size',
       'Active Mean', 'Idle Mean', 'Label'],
      dtype='object')
```

In [14]:
```
#Displays original label values
df_new['Label'].value_counts()
```

Out[14]:
```
Label
BENIGN          288548
Infiltration    199531
Name: count, dtype: int64
```

In [15]:
```
df_new
```

Out[15]:

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Mean	Bwd Packet Length Mean	Flow Bytes/s	Flow Packets/s	Packet Length Mean	Packet Length Std	Average Packet Size	Active Mean	Idle Mean	Label
0	53	62171	2	2	78	164	39.00	82.000000	3.892490e+03	64.338679	56.200000	23.552070	70.250000	0.00000	0.0	BENIGN
1	2710	48	2	0	4	0	2.00	0.000000	8.333333e+04	41666.666670	2.000000	0.000000	3.000000	0.00000	0.0	Infiltration
2	443	4792909	5	1	135	46	27.00	46.000000	3.776412e+01	1.251849	32.428571	18.866700	37.833333	0.00000	0.0	BENIGN
3	80	115596470	75	82	342	118155	4.56	1440.914634	1.025092e+03	1.358173	749.981013	946.124811	754.757962	22092.63636	10000000.0	BENIGN
4	2910	14	2	2	4	12	2.00	6.000000	1.142857e+06	285714.285700	3.600000	2.190890	4.500000	0.00000	0.0	BENIGN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
488110	53	238	2	2	86	228	43.00	114.000000	1.319328e+06	16806.722690	71.400000	38.888302	89.250000	0.00000	0.0	BENIGN
488111	8899	3	2	0	4	0	2.00	0.000000	1.333333e+06	666666.666700	2.000000	0.000000	3.000000	0.00000	0.0	Infiltration
488112	53	250	2	2	82	206	41.00	103.000000	1.152000e+06	16000.000000	65.800000	33.958799	82.250000	0.00000	0.0	BENIGN
488113	53	70581019	2	2	117	231	58.50	115.500000	4.930504e+00	0.056672	82.800000	30.817203	103.500000	41306.00000	70300000.0	BENIGN
488114	5679	121	2	2	4	12	2.00	6.000000	1.322314e+05	33057.851240	3.600000	2.190890	4.500000	0.00000	0.0	BENIGN

488079 rows × 16 columns

In [16]:
```
#Rename values in Label column
df_new['Label'] = df_new['Label'].replace({
    'BENIGN': 'Normal Traffic',
    'Infiltration': 'Attack'
})
```

In [17]:
```
#Displays label values after renaming
print("Label values after renaming:")
print(df_new['Label'].value_counts())
```

```
Label values after renaming:
Label
Normal Traffic    288548
Attack            199531
Name: count, dtype: int64
```

In [18]:
```
df_new
```

Out[18]:

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets	Total Length of Bwd Packets	Fwd Packet Length Mean	Bwd Packet Length Mean	Flow Bytes/s	Flow Packets/s	Packet Length Mean	Packet Length Std	Average Packet Size	Active Mean	Idle Mean	Label
0	53	62171	2	2	78	164	39.00	82.000000	3.892490e+03	64.338679	56.200000	23.552070	70.250000	0.00000	0.0	Normal Traffic
1	2710	48	2	0	4	0	2.00	0.000000	8.333333e+04	41666.666670	2.000000	0.000000	3.000000	0.00000	0.0	Attack
2	443	4792909	5	1	135	46	27.00	46.000000	3.776412e+01	1.251849	32.428571	18.866700	37.833333	0.00000	0.0	Normal Traffic
3	80	115596470	75	82	342	118155	4.56	1440.914634	1.025092e+03	1.358173	749.981013	946.124811	754.757962	22092.63636	10000000.0	Normal Traffic
4	2910	14	2	2	4	12	2.00	6.000000	1.142857e+06	285714.285700	3.600000	2.190890	4.500000	0.00000	0.0	Normal Traffic
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
488110	53	238	2	2	86	228	43.00	114.000000	1.319328e+06	16806.722690	71.400000	38.888302	89.250000	0.00000	0.0	Normal Traffic
488111	8899	3	2	0	4	0	2.00	0.000000	1.333333e+06	666666.666700	2.000000	0.000000	3.000000	0.00000	0.0	Attack
488112	53	250	2	2	82	206	41.00	103.000000	1.152000e+06	16000.000000	65.800000	33.958799	82.250000	0.00000	0.0	Normal Traffic
488113	53	70581019	2	2	117	231	58.50	115.500000	4.930504e+00	0.056672	82.800000	30.817203	103.500000	41306.00000	70300000.0	Normal Traffic
488114	5679	121	2	2	4	12	2.00	6.000000	1.322314e+05	33057.851240	3.600000	2.190890	4.500000	0.00000	0.0	Normal Traffic

488079 rows × 16 columns

In [19]:
```
# Select two numeric columns
data = df_new[['Flow Duration', 'Total Fwd Packets']]

# Summary statistics
summary_stats = data.agg(['sum', 'mean', 'std', 'skew', 'kurtosis'])

print(summary_stats)
```

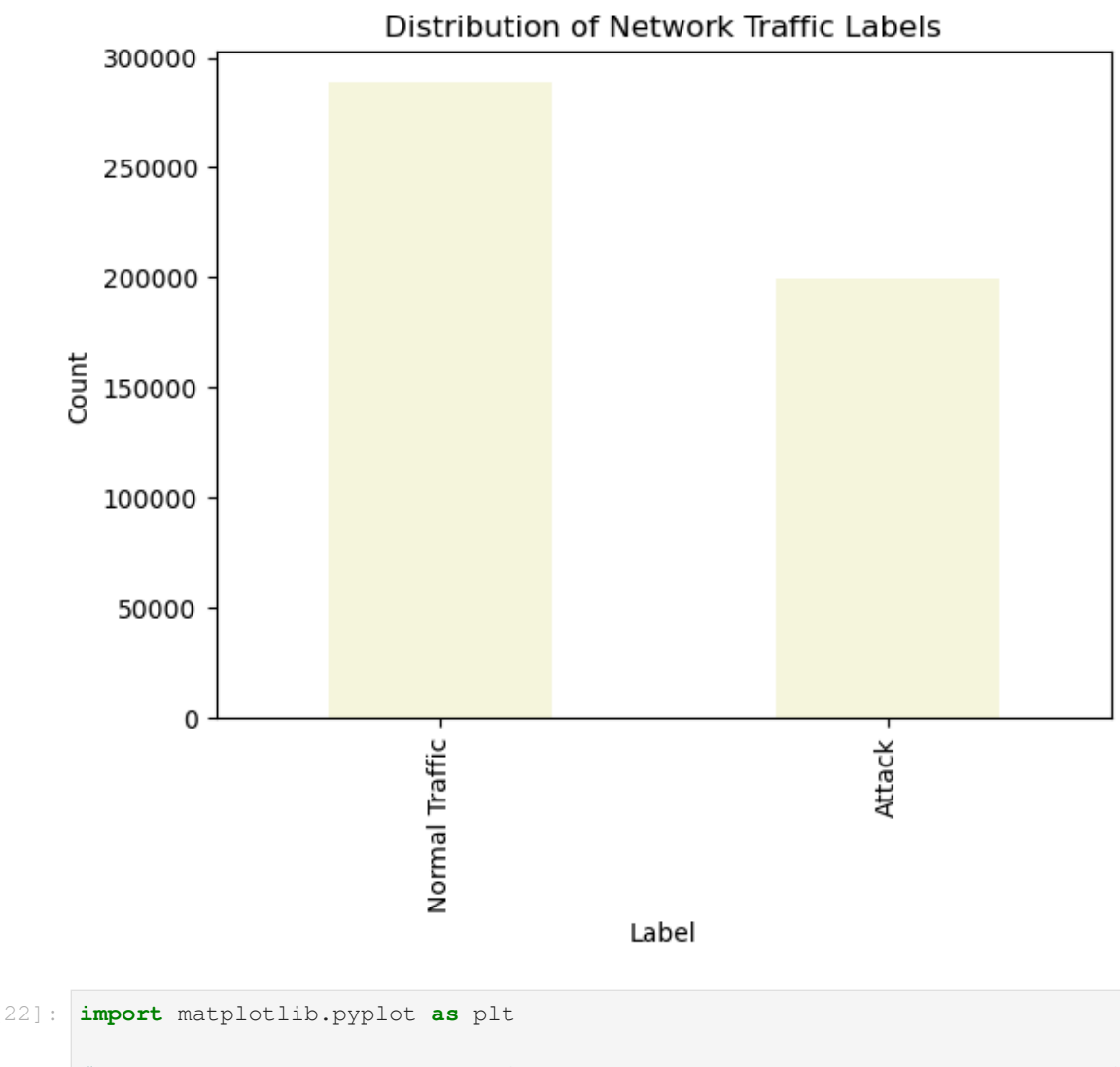
```
          Flow Duration  Total Fwd Packets
sum        4.364853e+12       3.029633e+06
mean       8.942923e+06       6.207259e+00
std        2.748182e+07       6.985526e+01
skew       3.250874e+00       1.370381e+02
kurtosis   9.246791e+00       3.040610e+04
```

In [20]:
```
# Compute correlation matrix for all numerical variables
correlation = df_new.corr(numeric_only=True)

# Flatten the matrix into pairs of features with correlation values
correlation = correlation.unstack()

# Exclude self-correlations (where a feature is correlated with itself)
correlation = correlation[correlation < 1]

# Sort correlations from highest to lowest to identify strongest relationships
correlation = correlation.sort_values(ascending=False)

# Display the top 5 strongest correlated feature pairs
print("Top 5 Strongest Correlation Pairs:")
print(correlation.head(5))
```

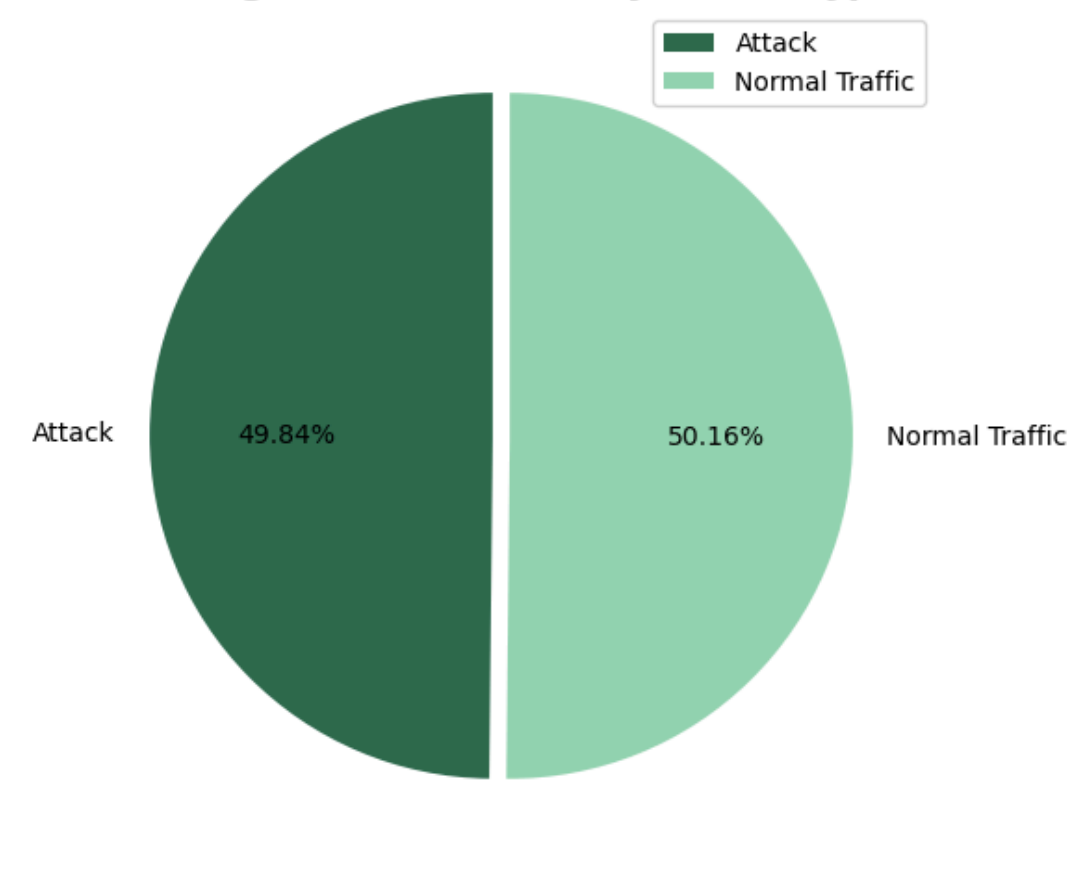
```
Top 5 Strongest Correlation Pairs:
Average Packet Size          Packet Length Mean             0.995068
Packet Length Mean           Average Packet Size            0.995068
Total Backward Packets       Total Length of Bwd Packets    0.961480
Total Length of Bwd Packets  Total Backward Packets         0.961480
Total Backward Packets       Total Fwd Packets              0.949848
dtype: float64
```

In [21]:
```
import matplotlib.pyplot as plt

# Count how many times each label appears in the Label column
counts_label = df_new["Label"].value_counts()

# Print the frequency counts to the console
print(counts_label)

# Create a bar chart using the label frequency counts
counts_label.plot(kind='bar', color='beige')

# Set the chart title and axis labels
plt.title("Distribution of Network Traffic Labels")
plt.xlabel("Label")
plt.ylabel("Count")

# display the result
plt.show()
```

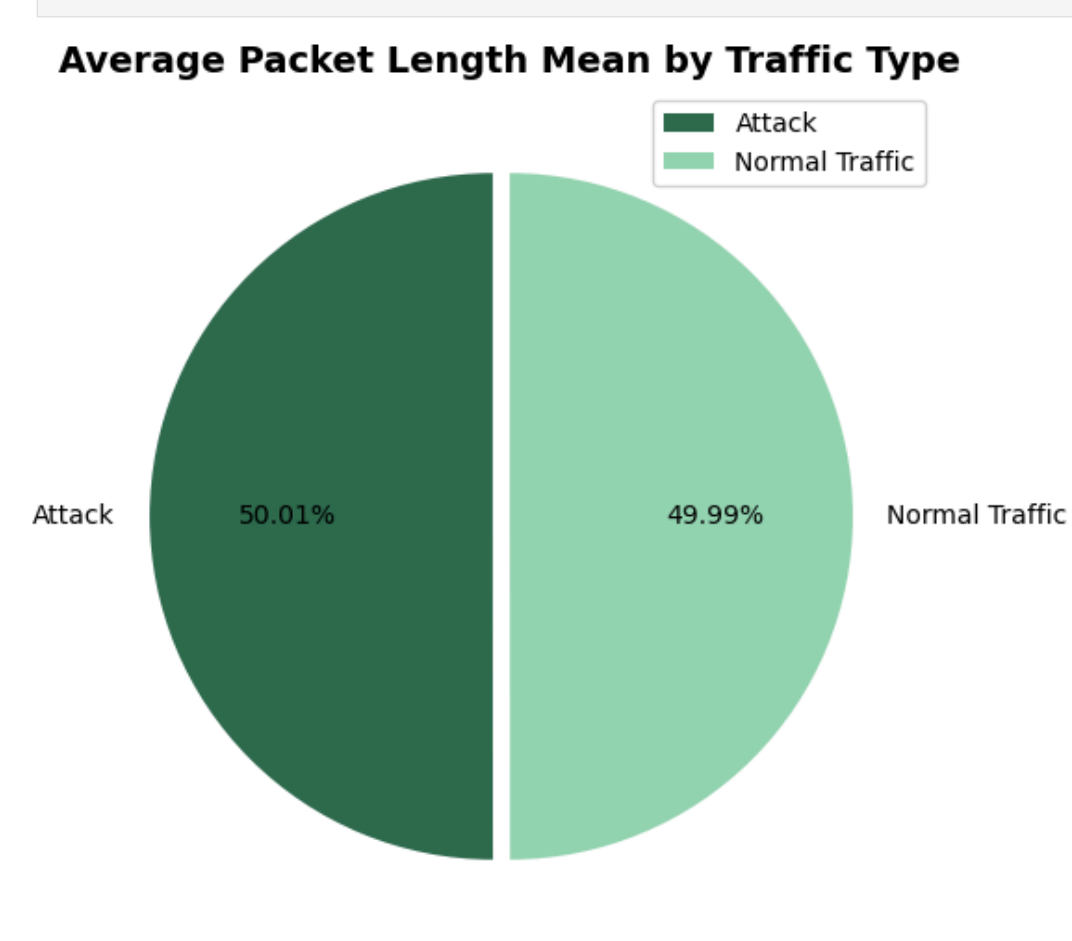
```
Label
Normal Traffic    288548
Attack            199531
Name: count, dtype: int64
```



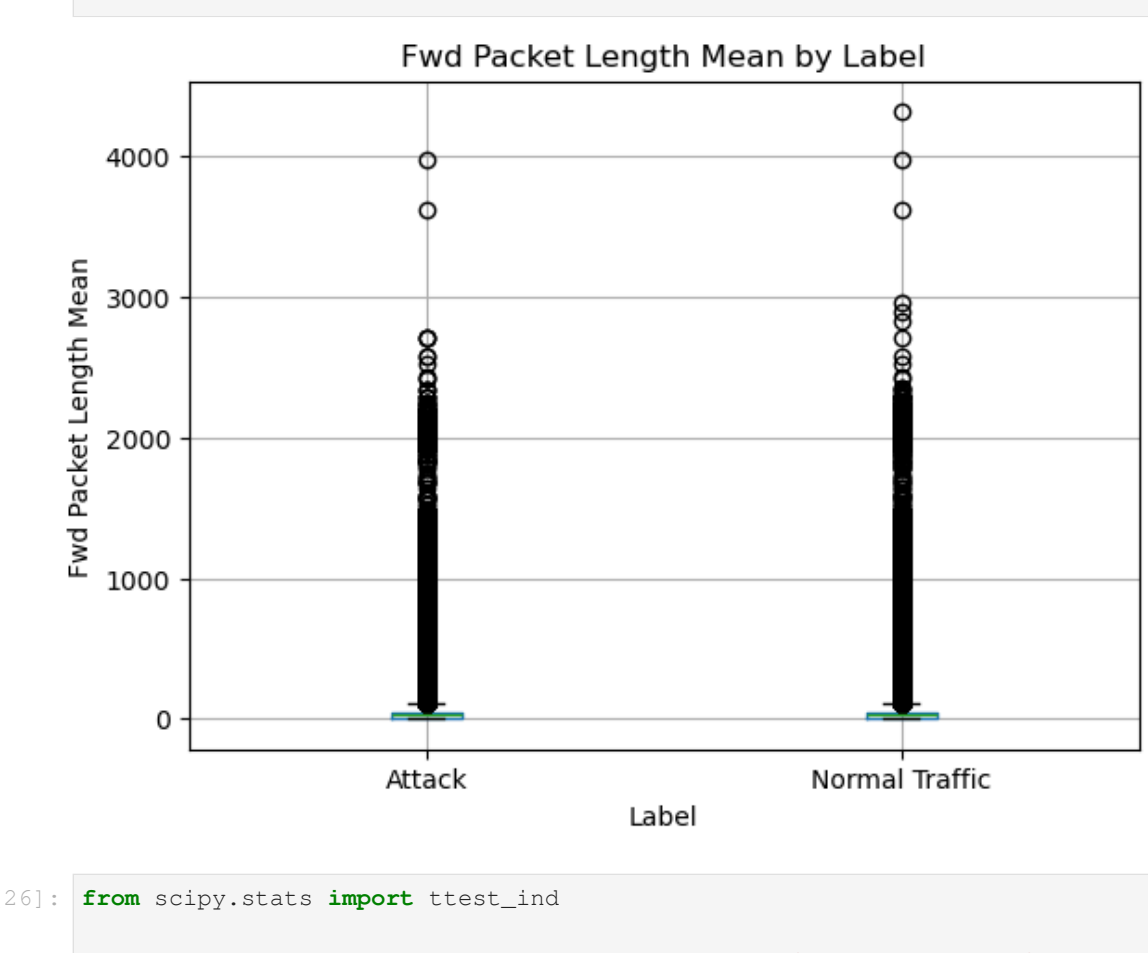
In [22]:
```
import matplotlib.pyplot as plt

# Calculate average Flow Duration and Packet Length Mean grouped by traffic label
grouped = df_new.groupby('Label')[['Flow Duration', 'Packet Length Mean']].mean()

# Initialise the pie chart with specified figure size
plt.figure(figsize=(6, 6))
plt.pie(grouped['Flow Duration'],
        labels=grouped.index,
        autopct='%1.2f%%',
        startangle=90,
        colors=['#2d6a4f', '#95d5b2'],
        explode=(0.05, 0)
       )

# Label the chart with a bold title and add a legend for reference
plt.title("Average Flow Duration by Traffic Type", fontsize=14, fontweight='bold')
plt.legend(grouped.index, loc='upper right')

# Show the final chart output
plt.show()
```



In [23]:
```
# Initialise the pie chart for Packet Length Mean with specified figure size
plt.figure(figsize=(6, 6))
plt.pie(grouped['Packet Length Mean'],
        labels=grouped.index,
        autopct='%1.2f%%',
        startangle=90,
        colors=['#2d6a4f', '#95d5b2'],
        explode=(0.05, 0))

# Label the chart with a bold title and add a legend for reference
plt.title("Average Packet Length Mean by Traffic Type", fontsize=14, fontweight='bold')
plt.legend(grouped.index, loc='upper right')

# Show the final chart output
plt.show()
```



In [24]:
```
# Create a boxplot comparing Fwd Packet Length Mean across traffic categories
df_new.boxplot(column="Fwd Packet Length Mean", by="Label")

# Set the chart title and suppress the default groupby subtitle
plt.title("Fwd Packet Length Mean by Label")
plt.suptitle("")

# Assign descriptive labels to both axes
plt.xlabel("Label")
plt.ylabel("Fwd Packet Length Mean")

# Render and display the boxplot
plt.show()
```

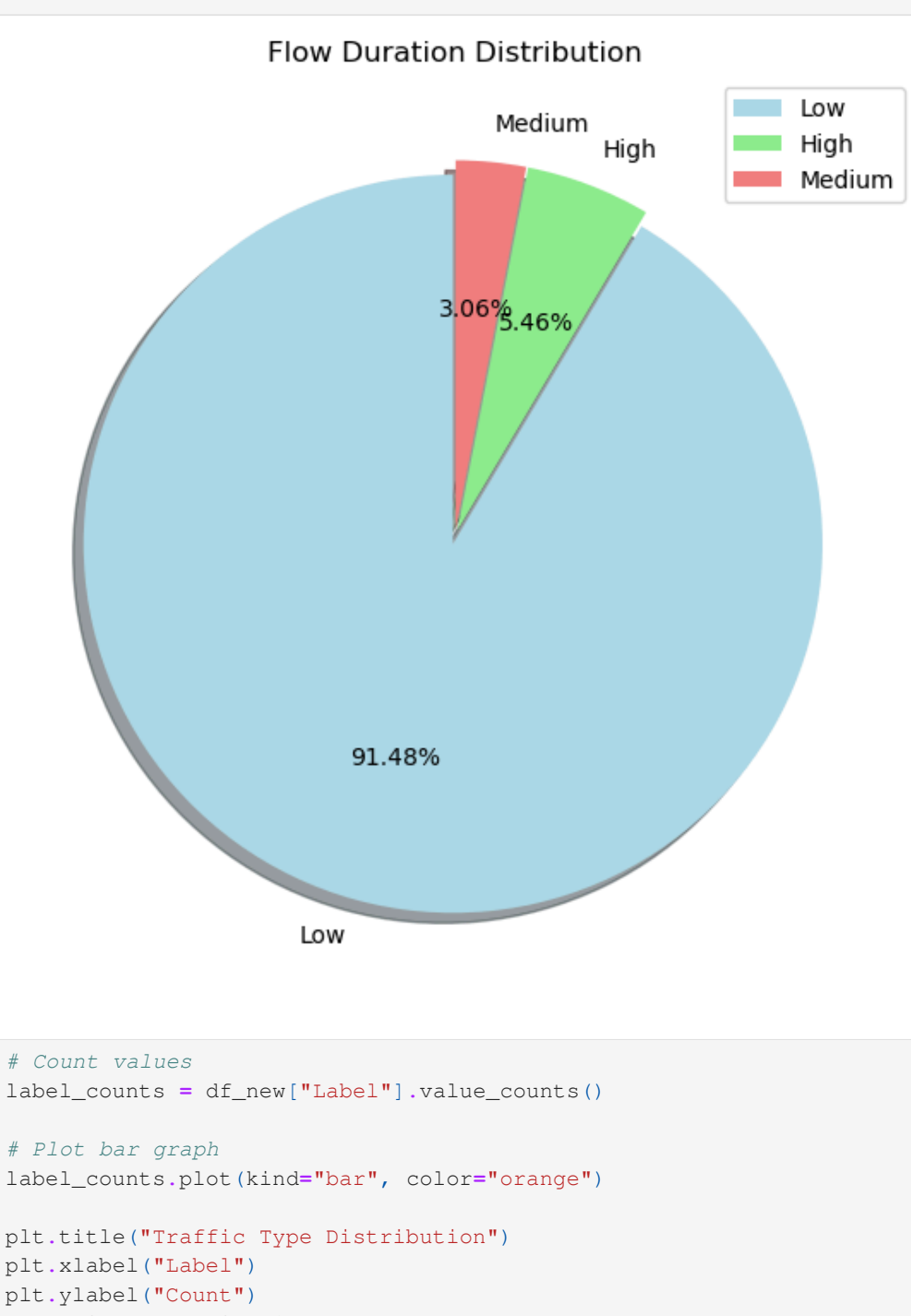


In [26]:
```
from scipy.stats import ttest_ind

normal = df_new[ df_new['Label'] == 'Normal Traffic']['Flow Duration']
attack =  df_new[ df_new['Label'] == 'Attack']['Flow Duration']

t_stat, p_value = ttest_ind(normal, attack)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

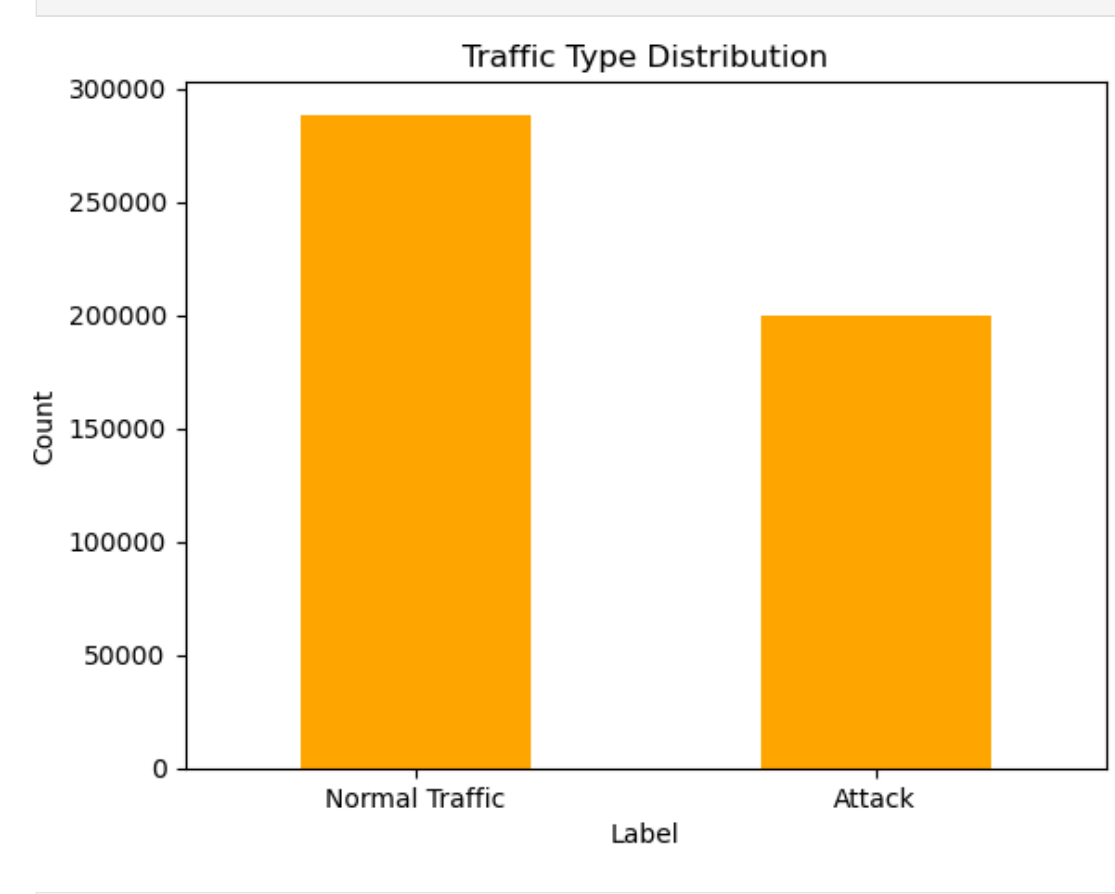
```
T-statistic: 0.7161327588416807
P-value: 0.473909720385436
```

In [27]:
```
#Appendix
```

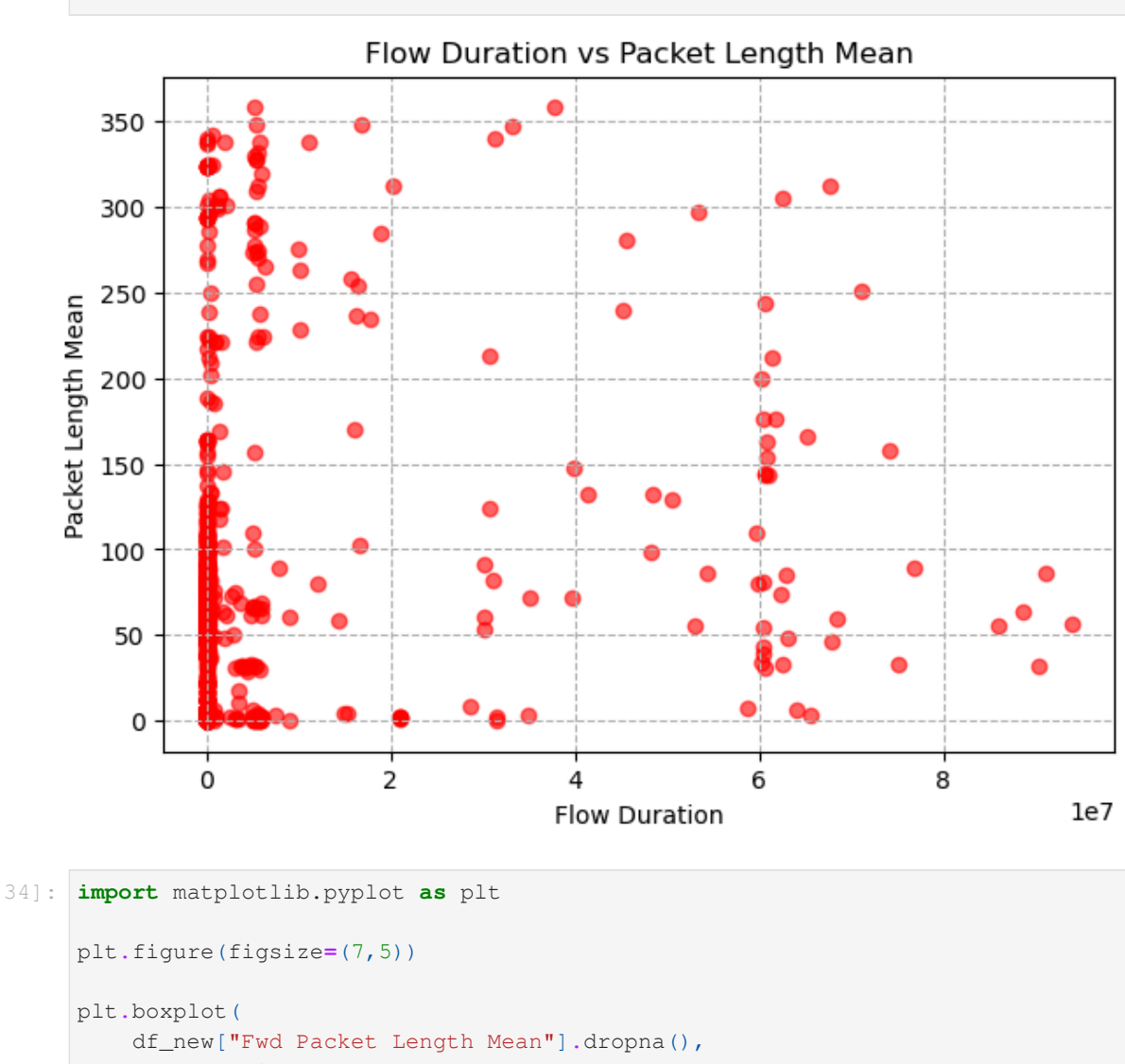
In [31]:
```
# Create categories
df_new["Flow Duration Category"] = pd.cut(
    df_new["Flow Duration"],
    bins=3,
    labels=["Low", "Medium", "High"]
)

# Count values
flow_counts = df_new["Flow Duration Category"].value_counts()

labels = flow_counts.index
sizes = flow_counts.values

colors = ["lightblue", "lightgreen", "lightcoral"]

plt.figure(figsize=(7,7))
plt.pie(
    sizes,
    labels=labels,
    colors=colors,
    autopct="%.2f%%",
    startangle=90,
    explode=[0.02, 0.02, 0.02],
    shadow=True
)

plt.legend()
plt.title("Flow Duration Distribution")
plt.show()
```



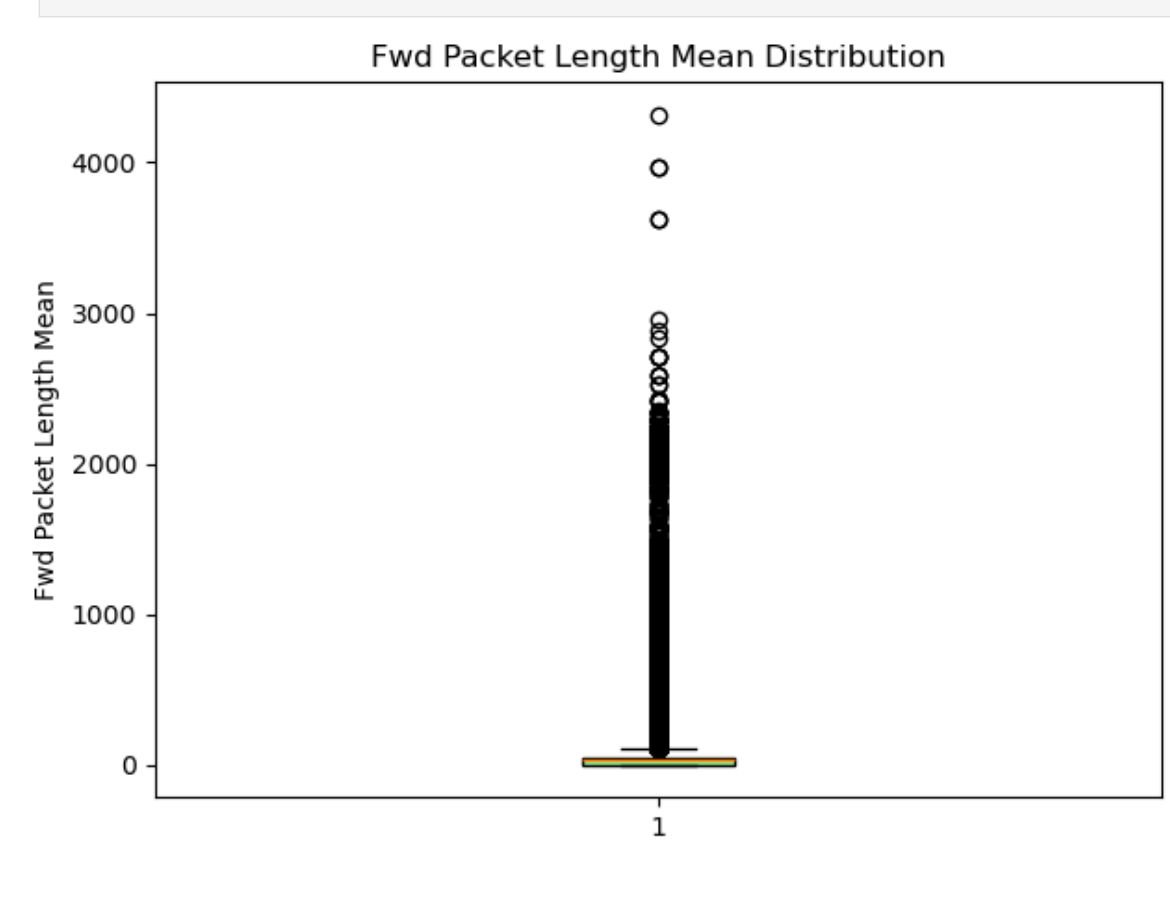
In [32]:
```
# Count values
label_counts = df_new["Label"].value_counts()

# Plot bar graph
label_counts.plot(kind="bar", color="orange")

plt.title("Traffic Type Distribution")
plt.xlabel("Label")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```



In [33]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(7,5))

# Select different columns
data = df_new[["Flow Duration", "Packet Length Mean"]].dropna()

# Remove outliers (95%)
data = data[
    (data["Flow Duration"] < data["Flow Duration"].quantile(0.95)) &
    (data["Packet Length Mean"] < data["Packet Length Mean"].quantile(0.95))
]

# Take sample for clarity
data = data.sample(1500)

# Scatter plot with different color
plt.scatter(data["Flow Duration"], data["Packet Length Mean"], color="red", alpha=0.6)

plt.title("Flow Duration vs Packet Length Mean")
plt.xlabel("Flow Duration")
plt.ylabel("Packet Length Mean")

plt.grid(True, linestyle="--")

plt.show()
```



In [34]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(7,5))

plt.boxplot(
    df_new["Fwd Packet Length Mean"].dropna(),
    patch_artist=True,
    boxprops=dict(facecolor="lightgreen")
)

plt.title("Fwd Packet Length Mean Distribution")
plt.ylabel("Fwd Packet Length Mean")

plt.show()
```



```
In [ ]:
```